

Human Position and Angle Detection System

AI/ML Internship — Day 8

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Module: Human Body Detection System

Introduction

Artificial Intelligence has significantly transformed the fashion industry through the development of intelligent image understanding systems. Modern applications such as virtual fitting rooms, online shopping assistants, automated product tagging, and fashion recommendation engines rely heavily on large-scale fashion datasets and robust image segmentation models.

DeepFashion is one of the most influential datasets developed for fashion analysis, while segmentation models provide the ability to identify and isolate clothing regions from complex images. This research focuses on understanding these technologies and their role in AI-powered fashion systems.

Objectives

The objectives of this research are:

- Study the DeepFashion dataset.
 - Understand image segmentation techniques.
 - Analyze modern segmentation architectures.
 - Explore clothing segmentation applications.
 - Investigate the role of segmentation in Virtual Try-On systems.
 - Perform a basic practical implementation.
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1. DeepFashion Dataset

1.1 Overview

DeepFashion is a publicly available large-scale fashion dataset developed to support computer vision research in the fashion industry. It contains thousands of images with detailed annotations that help AI systems recognize and understand clothing items.

The dataset is commonly used for:

- Clothing classification
- Fashion retrieval

- Garment detection
 - Virtual Try-On
 - Product recommendation
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1.2 Dataset Structure

The DeepFashion dataset contains several types of information:

Clothing Images

High-quality images containing various clothing styles.

Category Labels

Each garment belongs to a specific category such as:

- T-Shirt
- Shirt
- Jacket
- Jeans
- Dress
- Skirt

Bounding Boxes

Bounding boxes identify the approximate location of garments.

Landmark Annotations

Important garment points including:

- Collar
- Sleeve
- Waistline
- Hemline

Attribute Labels

Garments are annotated with properties such as:

- Color
 - Pattern
 - Fabric type
 - Sleeve length
-

1.3 Applications

E-Commerce

Automatic product categorization.

Fashion Search

Searching visually similar products.

Recommendation Systems

Suggesting personalized clothing items.

Virtual Dressing Rooms

Allowing users to digitally try garments.

Smart Inventory Management

Automated clothing organization.

1.4 Advantages

- Rich annotations
 - Large image collection
 - Supports multiple AI tasks
 - Widely used in research
 - Real-world fashion data
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1.5 Limitations

- Complex backgrounds
 - Some category imbalance
 - High storage requirements
 - Large computational cost
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2. Image Segmentation

2.1 Introduction

Image segmentation is a computer vision technique that divides an image into multiple meaningful regions.

Instead of assigning one label to the entire image, segmentation classifies each pixel individually.

This allows AI systems to distinguish clothing from body parts and background regions.

2.2 Types of Segmentation

Semantic Segmentation

Every pixel is assigned to a class.

Example:

- Shirt
- Pants
- Background

Instance Segmentation

Different objects belonging to the same class are separated individually.

Example:

Two jackets appearing in one image receive separate masks.

Panoptic Segmentation

A combination of semantic and instance segmentation.

Provides complete scene understanding.

3. Popular Segmentation Models

Fully Convolutional Network (FCN)

- Pixel-wise prediction
- Lightweight architecture
- End-to-end learning

Advantages

- Fast processing
- Simple design

Limitations

- Poor fine-detail segmentation

U-Net

Encoder-decoder architecture with skip connections.

Advantages

- High segmentation quality
- Excellent boundary preservation

Applications

- Medical imaging
 - Fashion segmentation
 - Object extraction
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Mask R-CNN

Performs:

- Object detection
- Classification
- Instance segmentation

It is widely used for clothing extraction.

DeepLabV3+

Uses Atrous Convolution to capture information at multiple scales.

Benefits

- High accuracy
 - Better garment boundaries
 - Strong contextual understanding
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SegFormer

A Transformer-based segmentation architecture.

Advantages:

- Lightweight
 - High performance
 - Better global feature extraction
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4. Clothing Segmentation in Virtual Try-On

The Virtual Try-On pipeline generally consists of:

User Image
↓
Human Detection
↓
Body Segmentation
↓
Clothing Segmentation
↓
Garment Mapping
↓
Virtual Try-On Generation
↓
Final Output

Segmentation plays a critical role because it allows the AI system to isolate clothing regions before applying garment transformations.

5. Practical Implementation

Garment Boundary Detection Using OpenCV

A simple implementation was developed to demonstrate the basic concept of clothing boundary extraction.

Python Code

```
import cv2

# Load image
image = cv2.imread("person.jpg")

if image is None:
    print("Image not found")
    exit()

# Resize image
image = cv2.resize(
    image,
    (500,700)
)
```

```
# Convert to grayscale
gray = cv2.cvtColor(
    image,
    cv2.COLOR_BGR2GRAY
)

# Apply threshold
_, mask = cv2.threshold(
    gray,
    120,
    255,
    cv2.THRESH_BINARY
)

# Save output
cv2.imwrite(
    "segmentation_output.jpg",
    mask
)

print("Segmentation Completed Successfully")

cv2.imshow(
    "Original Image",
    image
)

cv2.imshow(
    "Segmentation Output",
    mask
)
```

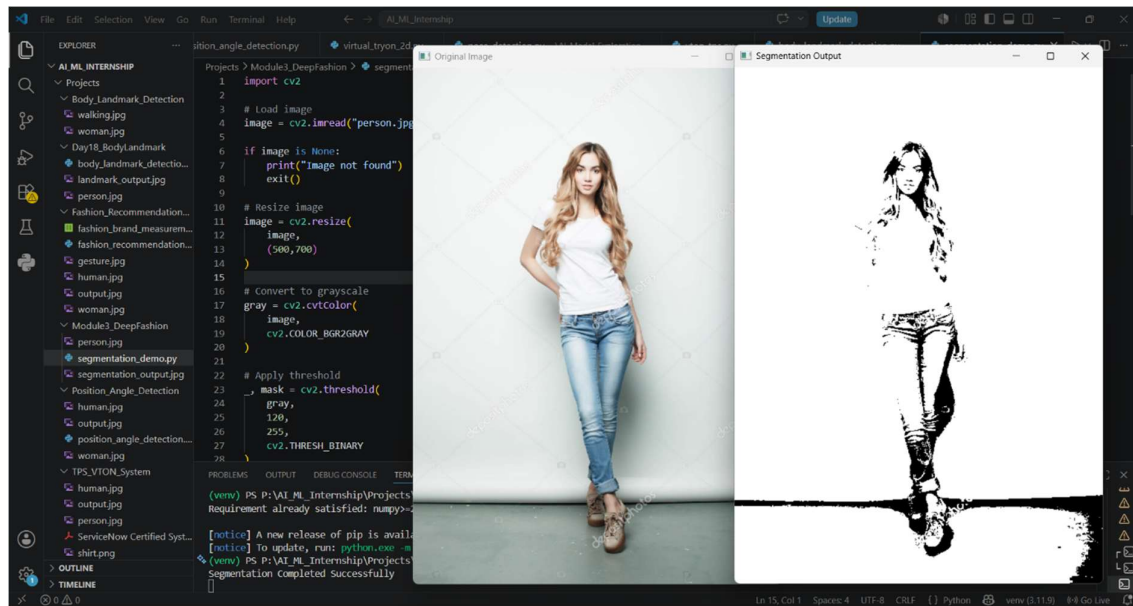
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Code Explanation

The program loads an image, converts it to grayscale, applies Gaussian Blur to reduce noise, and finally uses the Canny Edge Detection algorithm to highlight garment boundaries.

Although this implementation does not perform deep-learning-based segmentation, it demonstrates one of the basic concepts used in computer vision for identifying object contours.



6. Research Findings

- DeepFashion provides high-quality annotated fashion data.
- Segmentation models enable pixel-level garment understanding.
- Transformer-based architectures are becoming increasingly popular.
- Accurate segmentation improves Virtual Try-On realism.
- Fashion AI systems depend heavily on reliable segmentation techniques.

7. Learning Outcomes

During this research, the following concepts were studied:

- Fashion datasets
- Clothing annotations

- Image segmentation
 - Semantic segmentation
 - Instance segmentation
 - Deep learning architectures
 - Fashion AI applications
 - Virtual Try-On workflows
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8. Conclusion

This research explored the DeepFashion dataset and modern image segmentation technologies used in AI-powered fashion systems. DeepFashion provides comprehensive fashion data that supports model training, while segmentation algorithms enable precise garment extraction and scene understanding.

The integration of these technologies has significantly improved applications such as Virtual Try-On systems, intelligent retail platforms, and automated fashion analysis. Future developments in transformer-based architectures and large-scale datasets are expected to further enhance the capabilities of fashion AI solutions.
